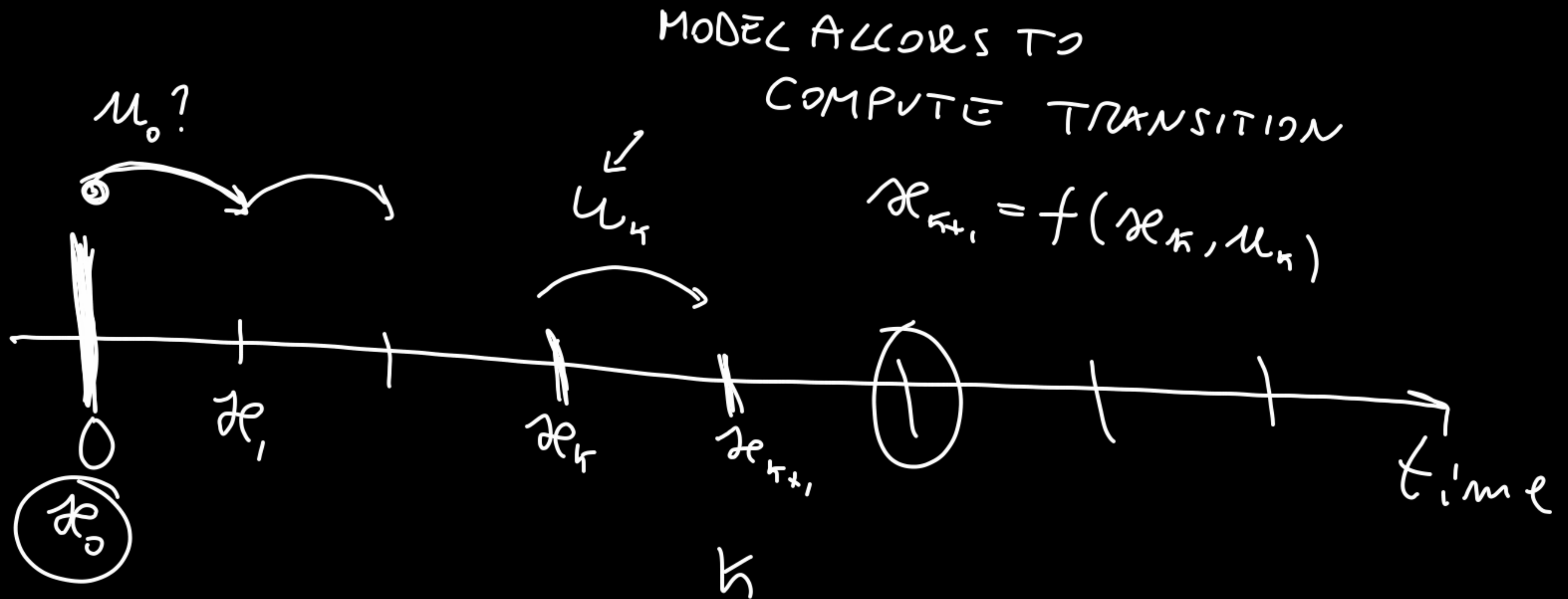


30/03/2026 - OPTIMALITY OF MPC - DUAL MODE CONTROL

- OPTIMALITY PRINCIPLE (BELLMAN PRINCIPLE).

- DYNAMIC PROGRAMMING $\leftrightarrow$ RL



ASSUMPTIONS:

- MODEL KNOWN AND PERFECT
- STATE MEASURABLE
- NO DISTURBANCES

∞ -HORIZON CONSTRAINED PROBLEM (0 IS THE INITIAL TIME)

$$\min_{\{u_0, u_1, u_2, \dots\}} \left\{ V^\infty = \sum_{k=0}^{\infty} \underbrace{\ell(x_k, u_k)}_{\text{STAGE COST}} \right\}$$

s.t.

IMPOSSIBLE
TO SOLVE:

— A MANY VARIABLES
CONSTRAINTS.

— TYPE OF ℓ, f

$$\begin{cases} x_{k+1} = f(x_k, u_k), & k=0, 1, \dots \end{cases}$$

x_0 known

DYNAMICS

$$x_k \in X$$

$$u_k \in U$$

$$, k=0, 1, \dots$$

SIMPLE CASE: ℓ QUADRATIC, f LINEAR

CONSTRAINED, ∞ -HORIZON $\mathbb{Q} \mathbb{R}$ PROBLEM

$$\min_{u_0, u_1, \dots} \left\{ V = \sum_{k=0}^{\infty} \underbrace{\left(x_k^T Q x_k + u_k^T R u_k \right)}_{\ell(x_k, u_k)} \right\}$$

IMPOSSIBLE TO SOLVE

DIFFICULTY:

- CONSTRAINTS ON x_k, u_k .

s.t.

$$\begin{cases} x_{k+1} = A x_k + B u_k, & k=0, 1, 2, \dots \\ x_0 \text{ known} \end{cases}$$

$$x_k \in X$$

$$u_k \in U$$

$$k=0, 1, \dots$$

SIMPLIFY $\rightarrow$ UNCONSTRAINED, ∞ -HORIZON LQR PROBLEM:

$$\min_{\{u_0, u_1, \dots\}} \left\{ V = \sum_{k=0}^{\infty} (x_k^T Q x_k + u_k^T R u_k) \right\}$$

s.t.

CAN BE
SOLVED

$$x_{k+1} = A x_k + B u_k \quad k=0, 1, 2, \dots$$

x_0 KNOWN

FEEDBACK SOLUTION

$$u_k^* = K x_k^*$$

OPTIMAL CONTROL
GAIN MATRIX

⑥

$$K = - \left(\underset{\uparrow}{B^T} \underset{\uparrow}{W} \underset{\uparrow}{B} + \underset{\uparrow}{R} \right)^{-1} \underset{\uparrow}{B^T} \underset{\uparrow}{W} \underset{\uparrow}{A}$$

$$\underline{W = A^T W A + Q - A^T W B (B^T W B + R)^{-1} B^T W A} \quad \text{LYAPUNOV EQUATION}$$

$$u_k^* = K x_k^*$$

$$(u_0^*, u_1^*, u_2^*, \dots)$$

OPTIMAL (*) CONTROL SEQUENCE

$$(x_0^*, x_1^*, x_2^*, \dots)$$

OPTIMAL STATE SEQUENCE

(OPTIMAL) COST OF CONTROL?

CALCULATION OF OPTIMAL COST

COMPUTE CONTROL / STATE OPT. SEQUENCES

$$u_0^* = K x_0^*$$

$$x_1^* = A x_0^* + B K x_0^* = (A + BK) x_0^*$$

$$u_1^* = K x_1^* = \dots$$

$$x_2^* = A x_1^* + B \overbrace{K}^{u_1^*} x_1^* = (A + BK) x_1^* =$$

$$= (A + BK)^2 x_0^*$$

GENERALIZE



OPTIMAL STATE
SEQUENCE

$$\mathbf{x}_k^* = (A+BK)^k \mathbf{x}_0 \quad (\text{FUNCTION OF INITIAL STATE})$$

OPTIMAL CONTROL
SEQ.

$$\mathbf{u}_k^* = K \mathbf{x}_k^*$$

$$\mathbf{u}_k^* = K (A+BK)^k \mathbf{x}_0 \quad \left(\begin{array}{cc} // & // \\ // & // \end{array} \right)$$

EVALUATE THE OBJ. FUNCTION

$$\sum_{k=0}^{\infty} \left((\mathbf{x}_k^*)^T Q \mathbf{x}_k^* + (\mathbf{u}_k^*)^T R \mathbf{u}_k^* \right) =$$

$$= \sum_{k=0}^{\infty} \left(\mathbf{x}_0^T \left[(A+BK)^k \right]^T Q (A+BK)^k \mathbf{x}_0 + \overbrace{\mathbf{x}_0^T \left[(A+BK)^k \right]^T K^T R K (A+BK)^k}^{(\mathbf{u}_k^*)^T} \mathbf{x}_0 \right) =$$

$$= x_0^T \underbrace{\left(\sum_{k=0}^{\infty} \left(\dots \right) \right)}_W x_0 = x_0^T W x_0$$

FOR ∞ -HORIZON UNCONSTRAINED LQR:

OPTIMAL CONTROL

$$u_k^* = K x_k^*$$

OPTIMAL COST

$$x_0^T W x_0$$

(W FROM LYAPUNOV
EQ.)

∞ -HORIZON CONSTRAINED LQR

ASSUMPTION(S):

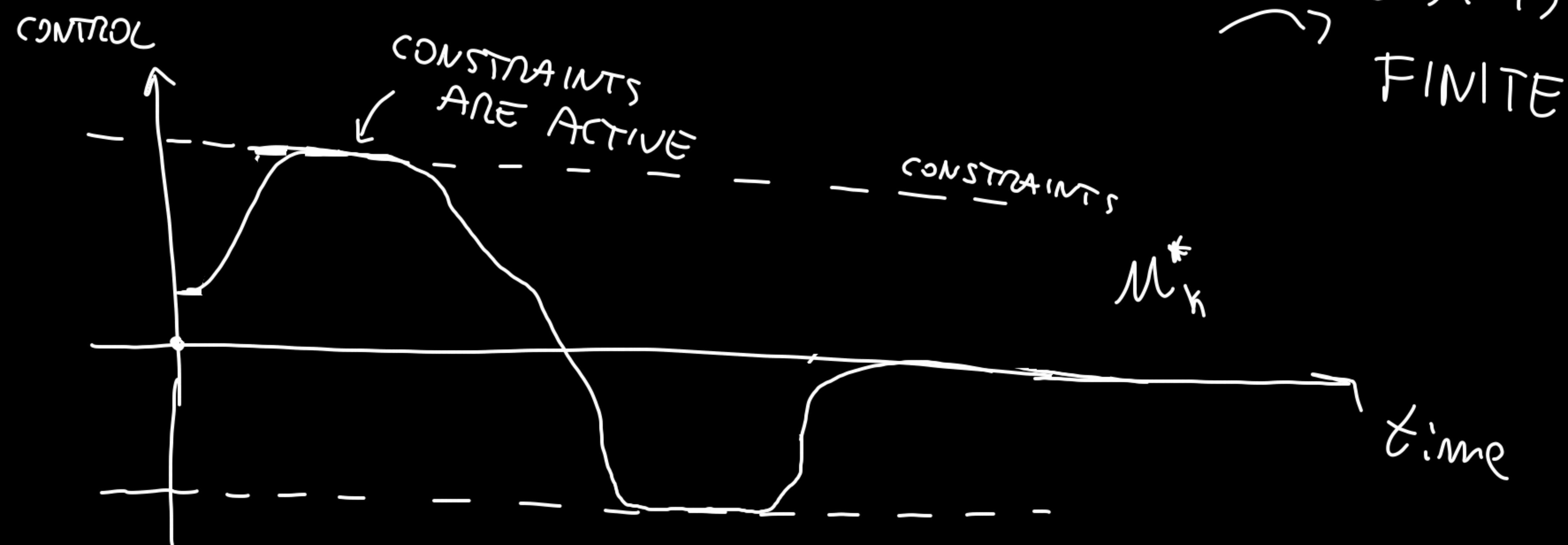
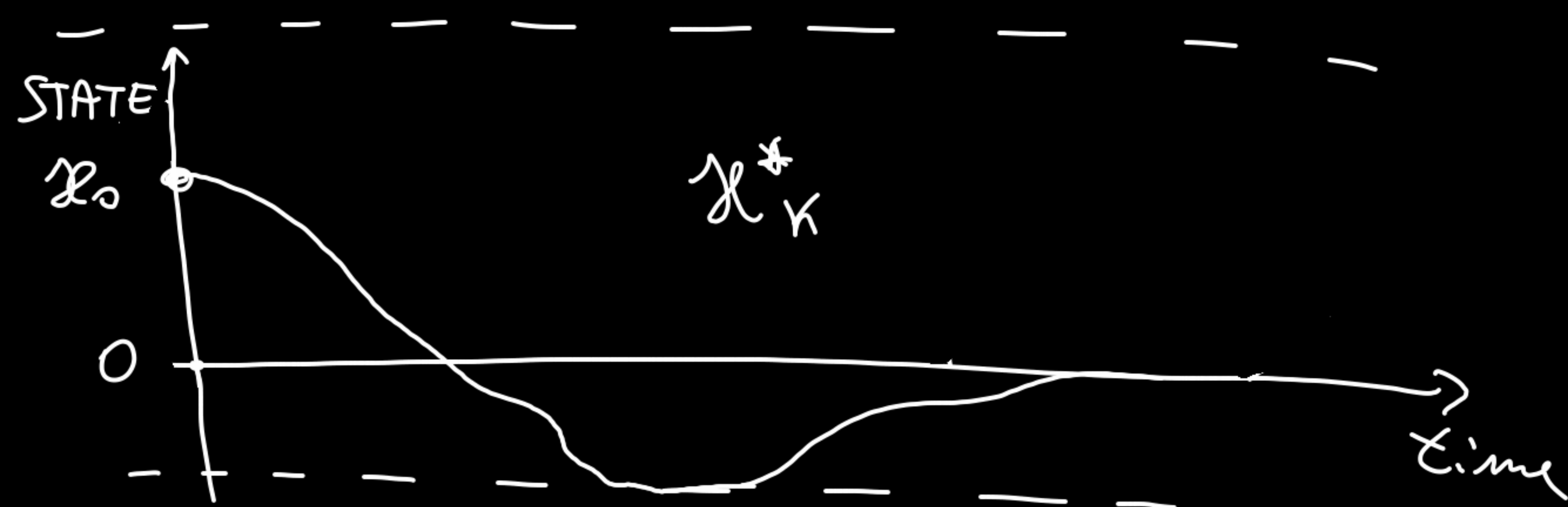
— SYSTEM IS CONTROLLABLE.

— $(x, u) = (0, 0)$ ARE
STRICTLY FEASIBLE



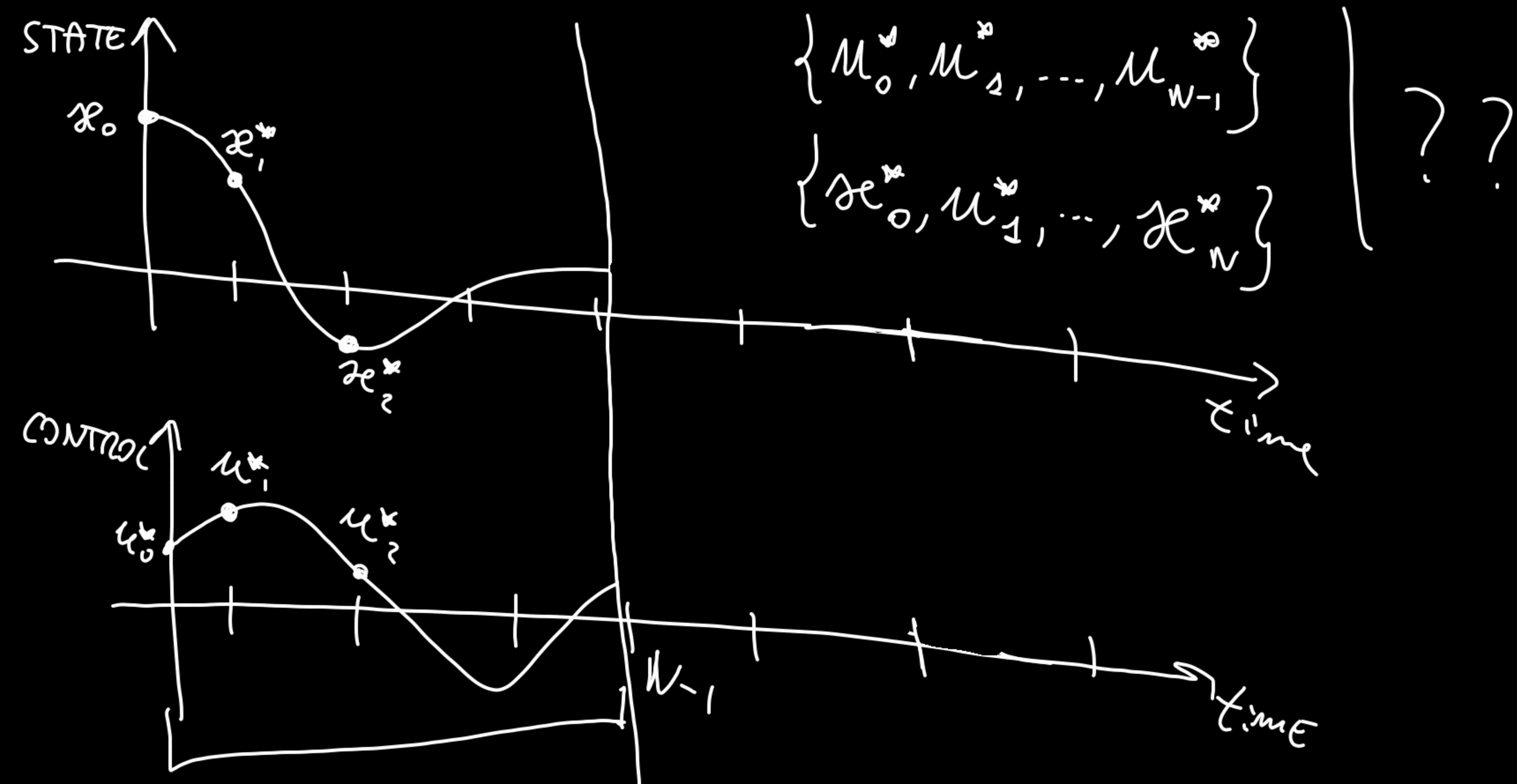
— $0 = f(0, 0)$

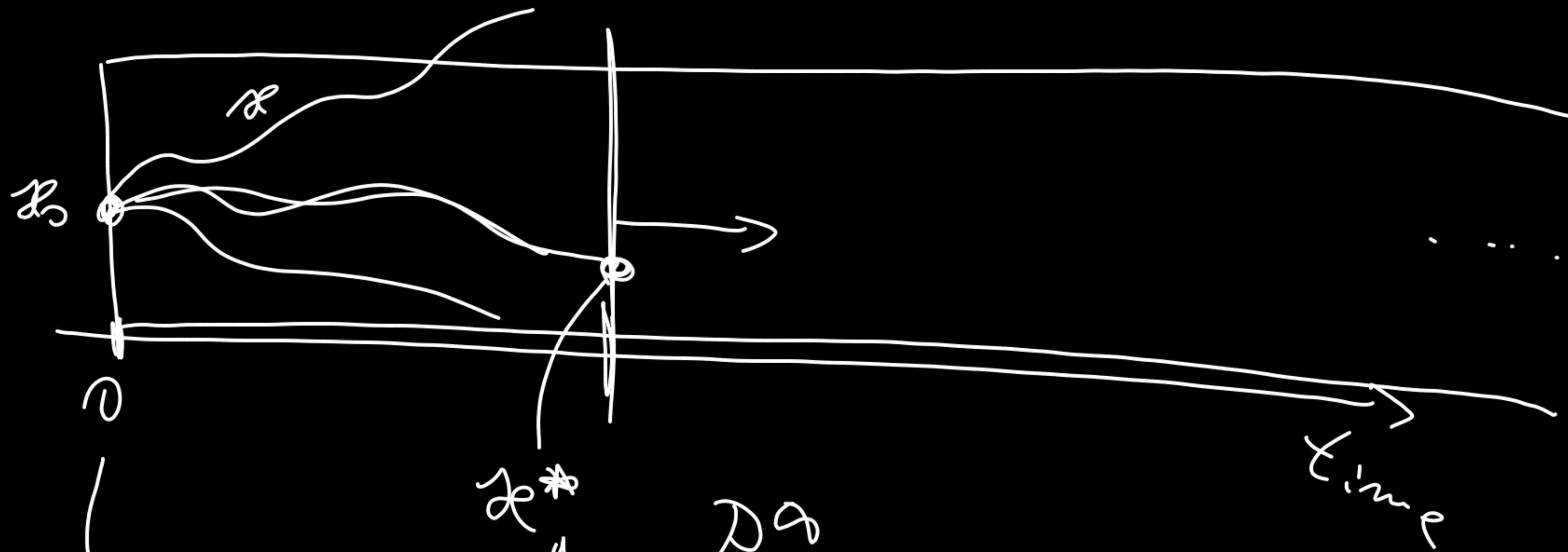
→ THERE IS AN
OPTIMAL
SOLUTION
(WE WANT TO
COMPUTE IT)



→ COST IS
FINITE

BELLMAN OPTIMALITY PRINCIPLE



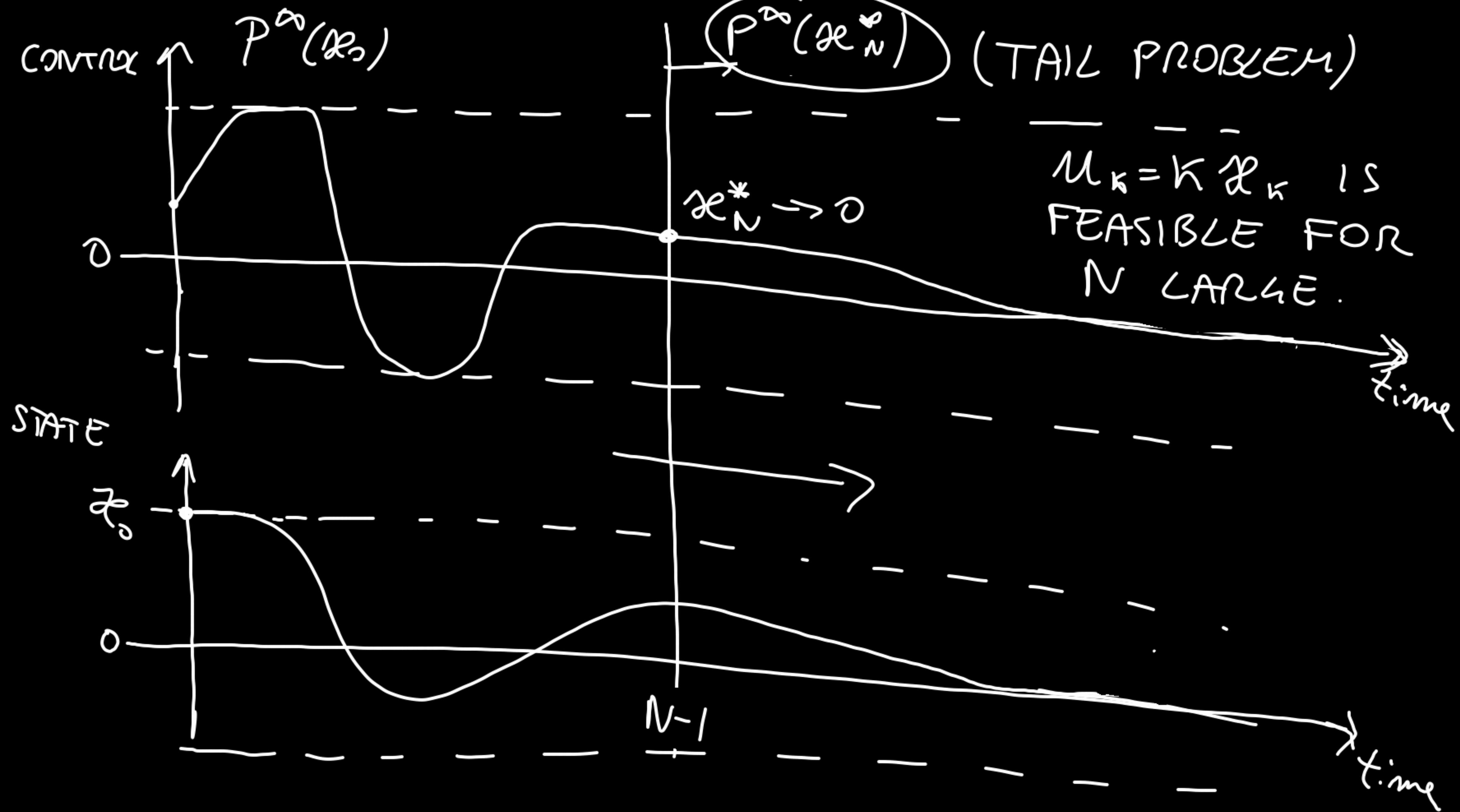


$$p_a(x_0)$$

$$x_n^*$$

$$p_a(x_n^*)$$

USE OF OPT. PRINCIPLE IN CONSTRAINED ∞ -HORIZON LQR



FACT:

$u_k = Kx_k$ (^{∞ -HORIZON} OPTIMAL SOLUTION OF UNCONSTRAINED (QR))

IS FEASIBLE FOR THE CONSTRAINED PROBLEM

STARTING FROM x_N^* , FOR N SUFF. LARGE



$u_k = Kx_k$ IS THE OPTIMAL SOLUTION FOR THE
TAIL PROBLEM

FOR N SUFF. LARGE:

• FIRST PART $\{u_0^*, u_1^*, \dots, u_{n-1}^*\}$ NOT KNOWN YET.

• TAIL $\{u_n^*, u_{n+1}^*, \dots\}$ IS GIVEN BY $u_k^* = K x_k^*$

CONSTRAINED ∞ -HORIZON $\angle QR$.

$V(x_N)$ TAIL COST

$$\min \left\{ \sum_{k=0}^{N-1} (x_k^T Q x_k + u_k^T R u_k) + \sum_{k=N}^{\infty} (x_k^T Q x_k + u_k^T R u_k) \right\}$$

CONSIDER
N SUFF.
LARGE

s.t. x_0 known

$$x_{k+1} = Ax_k + Bu_k$$

$$u_k \in U$$

$$x_k \in X$$

$$k=0, 1, 2, \dots, N-1$$

$$x_N^T V x_N$$

TAIL
CONSTRAINTS

~~$$x_{k+1} = Ax_k + Bu_k$$~~

~~$$u_k \in U$$~~

~~$$x_k \in X$$~~

$$k=N, N+1, \dots$$

FINITE-HORIZON "EQUIVALENT" FORMULATION (LQR)

$$\min_{\{u_0, u_1, \dots, u_{N-1}\}} \left\{ \sum_{k=0}^{N-1} (x_k^T Q x_k + u_k^T R u_k) + \overbrace{x_N^T V x_N}^{V_+(x_N)} \right\}$$

N VARIABLES
↓
CAN BE

SOLVED!

st

x_0 KNOWN

$$x_{k+1} = A x_k + B u_k, \quad k = 0, 1, \dots, N-1$$

$$x_k \in X, \quad k = 0, 1, \dots, N$$

$$u_k \in U, \quad k = 0, 1, \dots, N-1$$

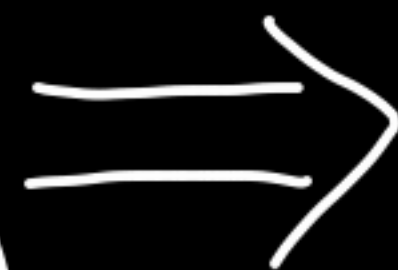
$V(x)$
VALUE FUNCTION

Δ -HORIZON P^∞ PROBLEM

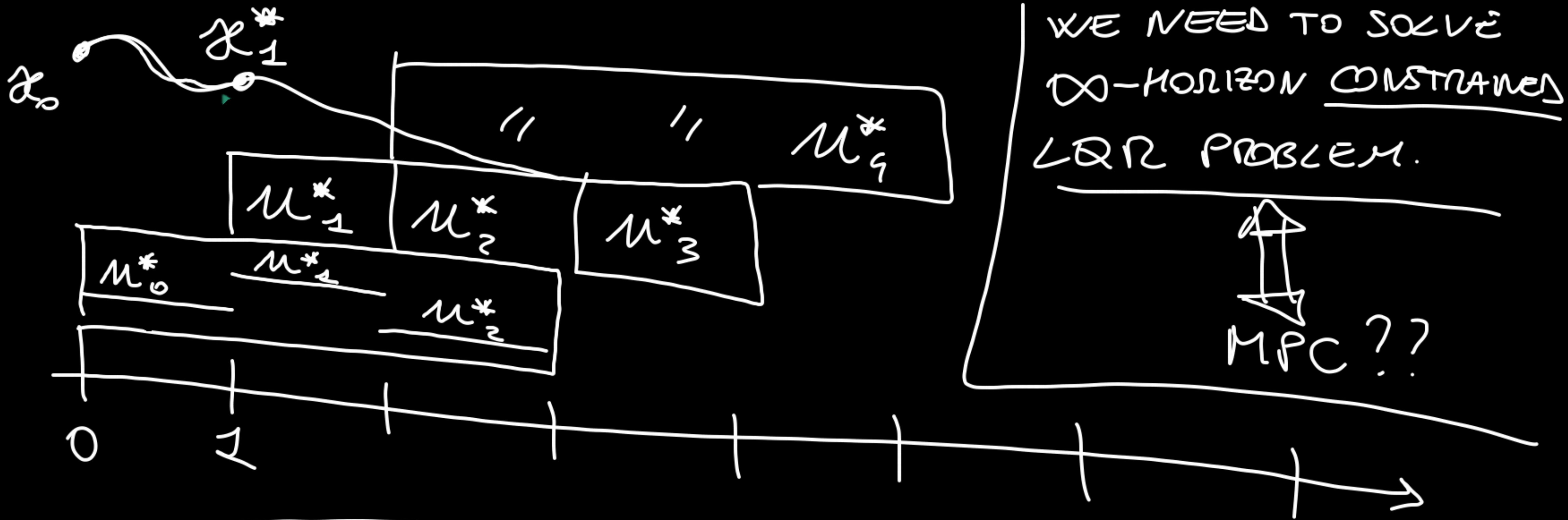
$u_0^{*\infty}, u_1^{*\infty}, u_2^{*\infty}, \dots$???

FINITE HORIZON PROBLEM

$u_0^{*N}, u_1^{*N}, \dots, u_{N-1}^{*N}$



$$u_k^{*N} = u_k^{*\infty}, \text{ FOR } k=0, 1, \dots, N-1$$



VALUE FUNCTIONS $V(x) : \mathbb{R}^n \rightarrow \mathbb{R}$
 COST OF OPTIMALLY CONTROLLING THE SYSTEM STARTING FROM x .
 For LQR $V(x) = x^T W x$